FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Cardiovascular Examination

Clinical skill | Week 2 · Diarrhoea, fluids, HIV & anaemia | Dr Aliku

A good paediatric cardiovascular examination is opportunistic and age-adapted. Inspect for growth, cyanosis, breathlessness and syndromes; feel all pulses (always the femorals) and the apex; listen in every area and at the back; then hunt for heart failure (in infants, a big liver and fast breathing) and finish with oxygen saturations in the right hand and a foot, BP and the growth chart, so you can tell an innocent murmur from heart disease.

Must know before the ward round

1. Approximate normal ranges: heart rate 110–160/min in newborns, 100–160 in infants, 80–140 at 2–5 years, 70–120 at 5–12 years and 60–100 over 12 years. WHO calls tachycardia >160 under 12 months and >120 at 1–5 years a sign of heart failure.
2. Fast breathing (WHO/IMCI): ≥60/min under 2 months, ≥50 at 2–12 months, ≥40 at 1–5 years. Lower limit of systolic BP (PALS): <60 mmHg in term neonates, <70 in infants, <70 + 2 × age in years at 1–10 years, <90 from 10 years.
3. Always palpate the femoral pulses. Weak or absent femorals, or radio-femoral delay, suggest coarctation of the aorta (or a duct-dependent left heart in a sick neonate). Check right arm vs leg BP: a gradient over 10–20 mmHg is abnormal.
4. Heart failure in infants looks like fast breathing, sweating and tiring with feeds, poor weight gain, tachycardia, a gallop and hepatomegaly. Oedema and a raised JVP are late or hard to see in infants.
5. Central cyanosis (tongue/lips) means SpO_2 is usually below 85% and is hard to see with anaemia or dark skin, so use a pulse oximeter. Newborn screen: pass if ≥95% in the right hand or foot with a difference ≤3%; <90% is a fail.
6. Innocent murmurs follow the 7 S's: Sensitive (changes with posture or breathing), Short, Single (no clicks), Small (localised), Soft (≤2/6), Sweet (not harsh), Systolic. The child must be asymptomatic, with normal pulses, normal S2 and normal growth.
7. Red flags for a pathological murmur: any diastolic murmur, pansystolic, grade ≥3/6 or a thrill, harsh quality, abnormal S2 (fixed split or single loud), clicks, symptoms, abnormal pulses or SpO_2, or a syndrome.
8. Grade systolic murmurs 1–6 (Levine): 4/6 means a thrill is present; 5/6 is heard with the stethoscope edge on the chest; 6/6 is heard with the stethoscope just off the chest.
9. Apex: 4th intercostal space just lateral to the midclavicular line in under-5s, 5th space at the midclavicular line in older children. Heaving (pressure load) vs thrusting/displaced (volume load); a left parasternal heave means RV hypertrophy.
10. Classic lesions: VSD = harsh pansystolic at the lower left sternal edge. ASD = ejection systolic at the upper left sternal edge + fixed split S2. PDA = continuous machinery murmur below the left clavicle + bounding pulses. Pulmonary stenosis = ejection systolic at the upper left sternal edge + click, to the back. Aortic stenosis = ejection systolic at the upper right sternal edge to the carotids, with a suprasternal thrill.
11. Fever, anaemia and anxiety cause flow murmurs, so re-examine when the child is well. Anaemia (Hb <7–8 g/dL) itself causes a soft ejection systolic murmur and can precipitate heart failure.
12. Complete the exam with SpO_2 (right hand + foot), 4-limb BP when indicated, liver span, lung bases, oedema, plotting weight and length, and an ECG, CXR and echocardiogram if the murmur is not clearly innocent.

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1. Purpose and Indications

Heart disease in Ugandan children is mainly **congenital heart disease (CHD)** (about 8–9 per 1,000 live births; VSD is the commonest), **rheumatic heart disease** in school-age children, **cardiomyopathies**, and **heart failure secondary to severe anaemia, severe pneumonia, malnutrition, AGN or sepsis**. You will find most of it with your eyes, fingers, a stethoscope and a **pulse oximeter**.

The paediatric cardiovascular examination differs from the adult one in three ways:

1. **It is opportunistic.** Listen to the heart **while the baby is quiet** (asleep on the mother's lap, feeding, sucking a finger). Do the upsetting parts (BP, femorals, liver) last.
2. **Signs of heart failure are different.** Infants with heart failure breathe fast, sweat, feed slowly and fail to gain weight. They have a **big liver**, not swollen ankles or a visible JVP.
3. **Age changes "normal".** Heart rate, breathing rate, BP and apex position all change with age, so you must know the normal ranges.

> CLINICAL PEARL
> The one thing never to forget: **feel the femoral pulses in every infant and measure SpO_2 in the right hand and a foot.** These two quick checks pick up coarctation and duct-dependent or cyanotic heart disease that a stethoscope alone can miss, and a newborn with these can collapse when the duct closes in the first 1–2 weeks.

Purpose: tell an **innocent murmur** from **heart disease**, recognise **heart failure** and **cyanosis**, and decide who needs an **ECG, CXR and echocardiogram** and referral.

Indications. Examine the cardiovascular system in every child admitted to hospital, and in detail when there is:

- a **murmur** found incidentally (e.g. during an illness or school screening);
- **breathlessness, sweating or tiring with feeds, poor weight gain**;
- **cyanosis**, low SpO_2, or "blue spells";
- **syncope, chest pain on exercise, palpitations**;
- **oedema, hepatomegaly**, or heart failure signs in severe anaemia, pneumonia, malnutrition or AGN;
- **fever with known heart disease** (endocarditis), or joint pains with a murmur (rheumatic fever);
- **syndromes** (Down, Turner, Noonan, Williams, Marfan, DiGeorge, congenital rubella, fetal alcohol);
- **maternal factors**: diabetes, rubella in pregnancy, lupus, drugs (valproate, lithium, warfarin), maternal CHD, HIV exposure (cardiomyopathy).

2. Contraindications and Precautions

- **There are no true contraindications.**
- Delay the non-urgent parts when the child is **critically ill**: in an emergency use the **ABCD (ETAT) approach** first and come back to the full examination once stable.
- **Avoid making a breathless or cyanosed infant lie flat or cry** (it worsens hypercyanotic spells and heart failure).
- **Privacy**: offer a **chaperone** for adolescents; keep a vest on older girls until you need to auscultate.
- **Correct BP cuff**: a cuff that is too small gives a falsely high reading.
- **Do not judge "no murmur" in a crying baby**: settle and listen again.

3. Relevant Anatomy and Normal Values

Normal values by age

Age	Heart rate (/min)	Respiratory rate (/min)	Systolic BP (mmHg)	Lower limit of SBP (PALS)
Newborn (0–28 days)	110–160	30–60 (WHO fast ≥60)	60–90	<60
1–12 months	100–160	30–50 (WHO fast ≥60 <2 mo, ≥50 2–12 mo)	70–100	<70
1–2 years	90–150	25–40 (WHO fast ≥40)	80–100	<70 + 2 × age
2–5 years	80–140	20–30 (WHO fast ≥40)	80–110	<70 + 2 × age
5–12 years	70–120	18–25	90–120	<70 + 2 × age (to 10 y), then <90
>12 years	60–100	12–20	100–130	<90

These are approximate bedside ranges (consistent with the Fleming *Lancet* 2011 centile review and APLS/PALS teaching). Fever raises HR by about 10/min per °C. **Sleeping** heart rates are lower; a **sleeping** tachycardia is significant (carditis, heart failure, anaemia, thyrotoxicosis). **WHO heart-failure tachycardia**: **>160/min under 12 months**, **>120/min at 1–5 years**.

Hypertension (AAP 2017): BP **≥95th centile** for age, sex and height on 3 occasions (from 13 years, ≥130/80). Use centile tables; a rough guide is that a systolic BP above about 110 in a toddler or 120 in a primary-school child needs repeating with the right cuff.

Other normal values

Item	Normal
Capillary refill	**≤2 s** (WHO/ETAT treats >3 s as abnormal)
BP cuff	Bladder **width ~40%** of mid-upper-arm circumference, **length 80–100%**
Arm vs leg BP	Leg BP equals or is higher than arm BP; an **arm-to-leg gradient >10–20 mmHg** is abnormal
Liver edge	Palpable **up to 2 cm** below the costal margin in infants (**1 cm** in older children)
Newborn pulse-oximetry screen	Pass **≥95%** in right hand or foot with a difference **≤3%**
Sinus arrhythmia	Normal in children (faster on inspiration)

Surface anatomy

- **Apex beat position**: **4th intercostal space just lateral to the midclavicular line** in children under about 5–7 years; **5th space at the midclavicular line** in older children. In **dextrocardia** it is on the right. Count down from the **sternal angle** (2nd space).
- **Auscultation areas**: **apex (mitral area)**; **lower left sternal edge (tricuspid area)**; **upper left sternal edge, 2nd space (pulmonary area)**; **upper right sternal edge, 2nd space (aortic area)**; plus the **left axilla**, **left infraclavicular area**, **neck/carotids** and **the back** between the scapulae.
- **Pre-ductal and post-ductal**: the **right hand** is pre-ductal; the **feet** are post-ductal. A difference shows right-to-left flow across the duct.
- **JVP**: between the heads of sternocleidomastoid with the child at 45° (older children only; the infant neck is too short).

Normal heart sounds in children

- **S2 splitting**: the aortic (A2) then pulmonary (P2) components split on **inspiration** and come together on expiration (physiological splitting). Best heard at the upper left sternal edge. **Fixed, wide splitting** = ASD. **Single S2** = one semilunar valve missing or very soft (pulmonary atresia, truncus, TGA where the aorta is anterior, severe TOF). **Loud P2** = pulmonary hypertension.
- **S3** is common and **normal** in healthy children and adolescents (low-pitched, early diastole, apex). With tachycardia and heart failure it forms a **gallop rhythm**.
- **S4** is always abnormal (stiff ventricle).

Transitional physiology that explains timing

- **Pulmonary vascular resistance** is high at birth and falls over the first weeks, so **left-to-right shunts** (VSD, AVSD, PDA) cause heart failure at **4–8 weeks**.
- The **ductus arteriosus** closes in the first **1–2 weeks**, so **duct-dependent** lesions present with cyanosis or shock then.

4. Equipment and Preparation

Equipment

Item	Why
Stethoscope with a paediatric/neonatal **bell and diaphragm** (an adult one works for older children)	Diaphragm for high-pitched sounds (S1, S2, most murmurs, AR); bell for low-pitched sounds (S3, S4, MS rumble), applied lightly
Pulse oximeter with paediatric/neonatal probe	Cyanosis is easy to miss clinically; pre-ductal (right hand) and post-ductal (foot) readings
BP cuffs in several sizes (neonatal, infant, child, small adult)	Bladder **width ~40% of mid-upper-arm circumference**, length **80–100%**. A small cuff gives a falsely high reading
Watch with a second hand	Count HR and RR for a full 60 seconds in infants
Tape measure, scale, length board/stadiometer, growth chart (WHO standards)	Failure to thrive is a key sign of heart disease; liver span
Torch/pen torch	Tongue and lips for cyanosis, conjunctivae, palate, teeth
Ophthalmoscope	Red reflex (congenital rubella cataract), Roth spots in endocarditis, hypertensive retinopathy
Urine dipstick	Haematuria in endocarditis; proteinuria in AGN/hypertension
Toys, a parent, warm room, privacy	A calm child gives a better examination; a chaperone for adolescents

Preparation, consent, positioning and exposure

1. **Introduce** yourself to the parent and child, confirm the child's name, age and the parent's relationship.
2. **Wash hands** (WHO 5 moments) and warm hands and stethoscope.
3. **Explain and get consent** from the parent and **assent** from the child; offer a **chaperone** for adolescents and respect privacy.
4. **Position**: infants on the parent's lap or lying supine; older children lying at **45°**.
5. **Exposure**: expose the chest fully (keep a vest for modesty in older girls until you need to auscultate).
6. Note whether the child is **calm or crying**; plan to auscultate first if the child is quiet.

5. Procedure Step by Step

The sequence follows **Hutchison's Clinical Methods (24th edition)**: general inspection (with vital signs and growth) → hands → pulses and BP → face → neck → precordium (inspection → palpation → auscultation) → signs of heart failure and completion. Adapt it for children: **auscultate first if the baby is quiet**, and do the upsetting parts (BP, femorals, liver) last if they make the child cry. Percussion of the heart is not routinely done.

Step 1. Introduction, consent and hand hygiene

As in section 4: introduce, confirm identity, wash and warm hands, explain, consent and assent, chaperone.

Step 2. Position and exposure

Infant on the parent's lap or supine; older child at 45°; chest exposed; room warm.

Step 3. General inspection ("end of the bed"), vital signs and growth

- **Well or unwell**; alert, irritable or lethargic.
- **Work of breathing**: rate, recession, nasal flaring, grunting, head bobbing, sitting forward.
- **Colour**: **central cyanosis** (tongue, lips), pallor, jaundice, mottling.
- **Sweating** (forehead during feeds), **feeding** (bottle/breast, how long it takes).
- **Dysmorphic features** suggesting a syndrome (table in section 6).
- **Nutrition and growth**: wasting, MUAC; check the child health card and **plot weight, length/height and head circumference** (WHO Z-scores).
- **Scars**: **median sternotomy** (open-heart surgery), **left lateral thoracotomy** (PDA ligation, coarctation repair, Blalock–Taussig shunt), right thoracotomy (shunt), small groin scars (cardiac catheter).
- **Medical equipment**: oxygen, NG tube (poor feeding in heart failure).
- **Chest shape**: precordial bulge (long-standing cardiomegaly), Harrison sulci, pectus.
- **Vital signs**: count the **RR for 60 seconds** while the child is calm; HR; temperature; **SpO_2 in the right hand and a foot** (or at completion if the child is upset by the probe).

Step 4. Hands

- **Temperature and capillary refill time**: press the **sternum (central CRT) or fingertip for 5 seconds**; normal **≤2 seconds** (WHO/ETAT treats **>3 s** as abnormal).
- **Clubbing** (loss of the nail-fold angle, Schamroth's window lost): cyanotic CHD (usually after 6–12 months of cyanosis), endocarditis, also lung disease (bronchiectasis, CF, TB) and chronic liver disease.
- **Peripheral cyanosis** (cold blue fingers; normal in a cold neonate), **palmar pallor** (anaemia).
- **Splinter haemorrhages, Osler nodes** (painful finger-pulp nodules), **Janeway lesions** (painless palmar macules): endocarditis.
- **Arachnodactyly** (Marfan), **single palmar crease** and clinodactyly (Down syndrome), **absent thumbs/radial ray defects** (Holt–Oram: ASD).

Step 5. Pulses and blood pressure

1. **Radial (older child) or brachial (infant)**: count **rate** for 60 seconds (15 seconds × 4 in an emergency), note **rhythm** (sinus arrhythmia, faster on inspiration, is normal in children), **volume** and **character**. Lift the arm to feel the "water-hammer" pulse in older children.
2. **Femoral pulses (always)**: both sides, at the mid-inguinal point. Compare with the right brachial/radial for volume and **delay**.
3. **Carotid** (older child): character (slow-rising in aortic stenosis); feel for a **carotid thrill** (AS).

4. **Blood pressure** with the right cuff: **right arm**, and **a leg** (calf or thigh) when coarctation is suspected or femorals are weak. **Four-limb BP** if the femorals are abnormal.

Step 6. Face, eyes and mouth

- **Eyes**: conjunctival pallor, jaundice; **red reflex** (rubella cataract); petechiae; Roth spots (fundoscopy).
- **Mouth**: **central cyanosis** of the tongue, **dental caries** (endocarditis risk), high-arched palate (Marfan), cleft palate (DiGeorge).
- **Facial features**: Down, Williams ("elfin" face), Noonan (hypertelorism, low-set ears, webbed neck), DiGeorge.

Step 7. Neck

- **JVP** in older children (sit at 45°, look between the heads of sternocleidomastoid; raised = right heart failure, fluid overload, tamponade). In infants the neck is too short; **use the liver** instead.
- Feel for a **suprasternal thrill** (aortic stenosis).
- Listen for **carotid bruits** and **venous hum** (innocent, above the clavicle).

Step 8. Precordium: inspect and palpate

1. **Inspect**: visible apex, precordial bulge, pulsations (left parasternal heave), scars.
2. **Palpate the apex beat**: count down the intercostal spaces from the sternal angle (2nd space) and describe the **position** (space and relation to the midclavicular line) and **character** (section 6).
3. **Left parasternal heave** (heel of the hand at the left sternal edge): right ventricular hypertrophy.
4. **Thrills** (palpable murmur = grade ≥4/6): flat of the hand at the apex, left sternal edge, upper left and right sternal edges, and the **suprasternal notch** (AS).

Step 9. Auscultation

1. **Listen in a sequence** with the **diaphragm, then the bell**:
   - **Apex (mitral area)**;
   - **Lower left sternal edge (tricuspid area)**;
   - **Upper left sternal edge, 2nd space (pulmonary area)**;
   - **Upper right sternal edge, 2nd space (aortic area)**;
   - then the **left axilla** (MR radiation), **left infraclavicular area** (PDA), **neck/carotids** (AS), and **the back** between the scapulae (coarctation, pulmonary stenosis/branch stenosis, PDA).
2. **Time the cardiac cycle**: in older children palpate the **carotid** (S1 just before, systole with the pulse); in infants, use the **brachial** or **femoral** pulse.
3. **Heart sounds**: S1 (loud in MS), S2 (**splitting at the upper left sternal edge with breathing**; single, fixed or loud P2), **added sounds** (S3 gallop, ejection click at the upper sternal edge in pulmonary or aortic stenosis/bicuspid valve, mid-systolic click in mitral valve prolapse, opening snap in MS).
4. **Murmurs**: describe each one systematically ("SCRIPT", section 11): **timing** (systolic: ejection vs pansystolic; diastolic: early vs mid; continuous), **site** of maximum intensity, **grade** (1–6), **radiation**, **pitch and quality** (harsh, blowing, rumbling, vibratory/musical), and **variation with posture or breathing**.
5. **Manoeuvres** (older children): **left lateral position with the bell** at the apex (MS, Carey Coombs murmur); **sitting forward in held expiration** at the left sternal edge (AR, pericardial rub); **sitting up then lying** (a Still's murmur gets louder lying down; a venous hum disappears when lying flat or when you press over the jugular vein).
6. **Pericardial rub**: scratchy, systolic and diastolic, changes with position.

Step 10. Look for heart failure and complete the examination

1. **Lung bases**: fine crackles (pulmonary oedema; but in infants, crackles are more often from pneumonia), effusions.
2. **Liver**: palpate from the right iliac fossa upwards; the liver edge is normally palpable **up to 2 cm** below the costal margin in infants (1 cm in older children). **Percuss the upper border** and measure the **span** (an enlarged edge may be a low-lying liver pushed down by hyperinflation). **Tender hepatomegaly** = right heart failure. **Pulsatile liver** = tricuspid regurgitation.
3. **Spleen**: endocarditis, malaria, haemoglobinopathy.
4. **Oedema**: **periorbital and sacral** in infants (and the scrotum or labia), ankles in older children; ascites.
5. **Oxygen saturation**: **right hand (pre-ductal) and a foot (post-ductal)**, if not already done.
6. **Growth chart**: plot weight, length/height and head circumference.
7. **Urine dipstick** (haematuria in endocarditis, protein in AGN), **temperature**, **fundoscopy** where indicated.
8. **Thank** the parent and child, cover the child, **wash hands**, and **summarise** your findings with a differential diagnosis and a plan for investigations (ECG, CXR, echo).

6. Findings and Interpretation

Pulses, apex and precordium

Finding	Mechanism / meaning	Think of
Bounding/collapsing pulse	Wide pulse pressure, diastolic run-off	PDA, aortic regurgitation, anaemia, sepsis, AV malformation
Weak/thready pulse	Low stroke volume	Shock, heart failure, severe aortic stenosis
Pulsus paradoxus	Volume falls on inspiration	Tamponade, severe asthma
Weak, delayed or absent femorals	Obstruction distal to the left subclavian	**Coarctation of the aorta** (or interrupted arch, or a duct-dependent left heart in a shocked neonate)
Slow-rising carotid, carotid or suprasternal thrill	Fixed LV outflow obstruction	Aortic stenosis
Arm-to-leg BP gradient >10–20 mmHg	Aortic obstruction	Coarctation
Apex displaced, thrusting	Volume load	VSD, PDA, MR, AR, cardiomyopathy
Apex heaving, sustained, not displaced	Pressure load	Aortic stenosis, coarctation, hypertension
Apex impalpable	Sound/impulse damped or heart elsewhere	Obesity, pericardial effusion, hyperinflation, dextrocardia (check the right side)
Left parasternal heave	Right ventricular hypertrophy	Pulmonary hypertension, pulmonary stenosis, TOF, large left-to-right shunt
Thrill	Murmur grade ≥4/6	Pathological murmur
Raised JVP (older child)	High right-sided pressure	Right heart failure, fluid overload, tamponade
Tender hepatomegaly	Hepatic congestion	Right heart failure (the infant's "JVP")
Pulsatile liver	Systolic backflow	Tricuspid regurgitation

Heart sounds

Finding	Meaning
Physiological S2 splitting	Normal
Fixed, wide split S2	ASD (equalised respiratory variation of RV volume)
Single S2	Pulmonary atresia, truncus, TGA, severe TOF
Loud P2	Pulmonary hypertension
S3 alone in a well child	Normal
S3 with tachycardia and heart failure	**Gallop rhythm**
S4	Always abnormal (stiff ventricle)
Ejection click (upper sternal edge)	Pulmonary or aortic stenosis, bicuspid valve
Mid-systolic click	Mitral valve prolapse
Loud S1, opening snap	Mitral stenosis

Grading murmurs (Levine)

Grade	Description
1/6	Very soft, heard only with concentration in a quiet room
2/6	Soft but easily heard
3/6	Moderately loud, **no thrill**
4/6	Loud, **with a thrill**
5/6	Heard with only the **edge of the stethoscope** on the chest
6/6	Heard with the **stethoscope just off the chest**

Diastolic murmurs are graded 1–4.

Innocent vs pathological murmurs

About **half to three-quarters of healthy children** have an innocent murmur at some time, often heard first during a fever.

Innocent murmur	Typical age	Features
Still's (vibratory) murmur	2–7 years	Short, **musical/vibratory** "twanging string", grade 1–3, **lower left sternal edge to apex**, louder lying down and with fever, softer when standing
Pulmonary flow murmur	School age, adolescents	Soft ejection systolic, upper left sternal edge, louder lying down and with fever, normal S2 splitting
Venous hum	3–8 years	**Continuous** humming, **below or above the clavicle** (usually right), **disappears lying flat, turning the head, or pressing on the jugular vein**
Peripheral pulmonary branch stenosis	Newborns, often preterm	Soft ejection systolic at the upper left sternal edge, **radiating to both axillae and the back**; resolves by about **6 months**
Supraclavicular/carotid bruit	Children, adolescents	Short systolic above the clavicle; softer with shoulder hyperextension

Innocent murmurs follow the 7 S's: Sensitive (changes with posture or breathing), Short, Single (no clicks), Small (localised), Soft (≤2/6), Sweet (not harsh), Systolic; in an asymptomatic, thriving child with normal pulses, S2 and SpO_2.

Signs that a murmur is pathological: symptoms (failure to thrive, breathlessness, sweating, cyanosis, syncope); **diastolic** or **pansystolic** timing; **grade ≥3/6** or a **thrill**; harsh quality; **loud at the upper left sternal edge with a click**; **abnormal S2** (fixed split, single, loud P2); **abnormal pulses** (weak femorals, bounding); **low SpO_2**; **hepatomegaly**; **syndrome**; murmur in a **neonate** (more likely to be structural).

Fever, anaemia and anxiety cause flow murmurs. **Anaemia (Hb <7–8 g/dL)** itself causes a soft ejection systolic murmur and can precipitate heart failure.

Classic lesions at the bedside

Lesion	Murmur	Other clues
VSD	**Harsh pansystolic**, **lower left sternal edge**, ± thrill; small VSD is loud, large VSD may be soft with mid-diastolic flow murmur at the apex	Heart failure at 4–8 weeks if large; loud P2 if pulmonary hypertension
ASD (secundum)	Soft **ejection systolic, upper left sternal edge** (increased pulmonary flow)	**Fixed, wide split S2**; usually asymptomatic in childhood
PDA	**Continuous "machinery" murmur, left infraclavicular** area	**Bounding pulses**, wide pulse pressure; preterm babies, congenital rubella
AVSD	Pansystolic (VSD and AV valve regurgitation)	**Down syndrome**, heart failure in infancy, superior QRS axis on ECG
Pulmonary stenosis	**Ejection systolic, upper left sternal edge**, radiating to the **back**; **ejection click**	RV heave, soft P2; Noonan
Aortic stenosis	**Ejection systolic, upper right sternal edge**, radiating to the **carotids**; **suprasternal thrill**; ejection click	Slow-rising pulse; syncope or chest pain on exercise; Williams (supravalvular)
Coarctation	Ejection systolic at the upper left sternal edge and **between the scapulae**	**Weak/delayed femorals**, arm hypertension, arm–leg BP gradient; Turner
Tetralogy of Fallot	**Ejection systolic, upper/mid left sternal edge** (RV outflow obstruction)	**Cyanosis**, **single S2**, clubbing later, **hypercyanotic spells**, squatting; boot-shaped heart on CXR
Mitral regurgitation (RHD)	**Apical pansystolic** to the **axilla**	Rheumatic fever history; displaced apex
Mitral stenosis (RHD)	**Apical mid-diastolic rumble** (bell, left lateral), **loud S1, opening snap**	AF, pulmonary hypertension
Aortic regurgitation	**Early diastolic decrescendo, left sternal edge** (sitting forward, expiration)	**Collapsing pulse**, wide pulse pressure

Syndromes and the heart

Syndrome	Typical heart lesion
Down syndrome (trisomy 21)	**AVSD** (atrioventricular septal defect), VSD, PDA, TOF; all babies with Down syndrome need an echo
Turner syndrome (45,X)	**Coarctation**, bicuspid aortic valve
Noonan syndrome	**Pulmonary stenosis** (dysplastic valve), hypertrophic cardiomyopathy
Williams syndrome	**Supravalvular aortic stenosis**, peripheral pulmonary stenosis; hypercalcaemia
DiGeorge (22q11 deletion)	**Conotruncal lesions**: TOF, truncus arteriosus, interrupted aortic arch

Syndrome	Typical heart lesion
Marfan syndrome	Aortic root dilatation, mitral valve prolapse
Congenital rubella	**PDA**, peripheral pulmonary stenosis (with cataract, deafness)
Fetal alcohol syndrome	VSD, ASD
Maternal diabetes	TGA, VSD, hypertrophic cardiomyopathy
Holt–Oram	ASD with upper-limb defects

Cyanosis and SpO_2

- **Central cyanosis** (tongue/lips) means SpO_2 is usually **below 85%** and is hard to see with anaemia or dark skin: use a pulse oximeter. **Peripheral cyanosis** alone is common in cold neonates.
- **SpO_2 low and does not rise with oxygen** → cyanotic CHD (TOF, TGA, tricuspid atresia, TAPVD, truncus). **Rises well** → lung disease.
- **Pre/post-ductal difference >3%** → duct-dependent systemic circulation or PPHN.

Bedside tests

Test	What it tells you
Pulse oximetry (right hand and foot)	Cyanosis; pre/post-ductal difference (duct-dependent systemic circulation, PPHN)
Newborn pulse-oximetry screen (after 24 h)	**Pass**: ≥95% in right hand or foot and difference ≤3%. **Fail**: <90% in either. **Repeat**: 90–94% or difference >3% (repeat once, then fail if still abnormal) (AAP 2020 algorithm)
Hyperoxia ("nitrogen washout") test (where available)	100% oxygen for 10 min: PaO_2 >150 mmHg (or big SpO_2 rise) suggests lung disease; PaO_2 <100 mmHg suggests cyanotic CHD
4-limb BP	Coarctation
HemoCue Hb	Anaemia (flow murmur, heart failure)
Blood glucose	Sick neonates, heart failure with poor feeding
Urine dipstick	Endocarditis (haematuria), AGN (blood + protein)

Laboratory tests and imaging

Test	When	Expected / abnormal finding	How it changes management
ECG	Any suspected structural disease, arrhythmia, syncope	Axis (superior axis in AVSD; right axis and RVH in TOF/PS), LVH, RVH, prolonged PR (rheumatic fever), arrhythmias, long QT	Supports the diagnosis; arrhythmia treatment. Hospitals and RRHs
Chest X-ray	Heart failure, cyanosis, suspected CHD	Cardiomegaly (cardiothoracic ratio >0.6 in infants, >0.5 in older children), **plethoric** vs **oligaemic** lung fields, **boot-shaped heart** (TOF), "egg on a string" (TGA), rib notching (coarctation, older children)	Classifies the lesion; excludes pneumonia. Most hospitals
Echocardiography with Doppler	**Any murmur not clearly innocent**, cyanosis, heart failure, Down syndrome, suspected RHD	Defines anatomy, shunt direction and size, gradients, ventricular function, pulmonary pressure	Definitive diagnosis and referral for surgery/catheter. **Uganda Heart Institute (Mulago)**, some RRHs and private/mission hospitals; handheld echo in RHD programmes

Test	When	Expected / abnormal finding	How it changes management
FBC	Pallor, heart failure, cyanosis	Anaemia; **polycythaemia** (Hb/Hct high in chronic cyanosis; iron deficiency is relative if MCV low)	Transfuse or treat iron deficiency (protects against hypercyanotic spells and stroke)
Blood cultures (×3)	Fever with heart disease	Organism	Endocarditis treatment
ESR/CRP, ASO	Suspected rheumatic fever	Raised	Jones criteria
U&E, creatinine	Heart failure on diuretics/ACE inhibitors	K^+, renal function	Dose adjustment
HIV test	Cardiomyopathy, pericardial effusion	Positive	ART; TB pericarditis screen

Worked example: innocent or not?

Child	Findings	Verdict
A, 4 y, febrile with malaria	Grade 2/6 vibratory systolic at LLSE, louder supine; normal pulses, S2 and SpO_2; growing well	**Innocent (Still's)**, accentuated by fever. Re-examine when well
B, 6 wk	Grade 2/6 systolic at upper left sternal edge radiating to both axillae and back; thriving	**Peripheral pulmonary branch stenosis** (innocent); review at 6 months
C, 2 y	Grade 2/6 ejection systolic at the upper left sternal edge, **fixed split S2**	**Pathological: ASD.** Echo
D, 8 wk	No murmur heard while crying; RR 70, sweating, liver 4 cm, poor weight gain	**Heart failure**: examine again when settled; echo (large VSD can be quiet)
E, 11 y	Early diastolic murmur at left sternal edge, collapsing pulse	**Pathological: aortic regurgitation** (RHD likely)

7. Clinical Acumen

First 5 minutes

1. **Is the child stable?** ETAT ABCD: airway, breathing (RR, recession, SpO_2), circulation (CRT, pulse volume, HR, cold hands), consciousness, dehydration.
2. If **shocked**: oxygen, IV/IO access, glucose; in a neonate with weak femorals or cyanosis, think **duct-dependent CHD** (cautious fluids 10 mL/kg, prostaglandin if available, urgent senior). In an older child with heart failure, **avoid large fluid boluses**.
3. If **cyanosed**: SpO_2 in the right hand and a foot; give oxygen and note the response.
4. If **in heart failure**: sit up, oxygen if SpO_2 <90% or distressed, furosemide 1 mg/kg, check Hb.
5. If **hypercyanotic spell**: knee-chest position, calm the child, oxygen, morphine (section 8).

Focused history: what to ask before you examine

Ask about	Why it matters
Feeding: duration of a feed (>20–30 min), stopping to breathe, **sweating with feeds**, vomiting	Heart failure in infants (large VSD, AVSD, PDA)
Breathing: fast breathing at rest, recession, recurrent "pneumonias"	Pulmonary overcirculation; heart failure

Ask about	Why it matters
Weight gain: crossing centiles down on the child health card	Failure to thrive from high energy use and poor intake
Colour: blue lips, "blue spells" when crying or feeding, squatting after exercise	Cyanotic CHD; **hypercyanotic (tet) spells**
Exercise tolerance: keeping up with peers, **syncope or chest pain on exertion**, palpitations	AS, HOCM, arrhythmia, pulmonary hypertension, RHD
Fever, joint pains, sore throat, skin sores, abnormal movements	Rheumatic fever, endocarditis
Swelling of eyes, legs, abdomen	Heart failure, nephrotic syndrome, AGN, malnutrition
Pregnancy history: maternal diabetes, rubella, fever, drugs, alcohol, HIV, lupus; prematurity	CHD risk; PDA in preterm; congenital rubella (PDA, PS)
Neonatal history: SpO_2 screen, murmur at discharge, collapse at 1–2 weeks	Duct-dependent lesions present when the duct closes
Family history: CHD, sudden death, cardiomyopathy, RHD	Recurrence risk (2–5% for a sibling); inherited conditions
Immunisation (UNEPI): measles-rubella, PCV, Hib	Rubella prevention; infection risk
HIV status and ART	HIV cardiomyopathy, pulmonary hypertension, endocarditis
Development	Syndromes; chronic heart failure delays motor milestones

Red flags: stop and escalate

RED FLAG

- **Neonate with cyanosis, SpO_2 <90%** (or pre/post-ductal difference >3%), **weak femorals**, **shock or acidosis in week 1–2**: suspect a **duct-dependent lesion**. Needs urgent senior review, prostaglandin if available, and referral (Uganda Heart Institute).
- **Heart failure in infancy**: fast breathing, sweating with feeds, gallop, hepatomegaly, poor weight gain.
- **Hypercyanotic spell**: deepening cyanosis, irritability, hyperpnoea, a **murmur that softens or disappears** (less flow across the RVOT), floppiness.
- **Syncope or chest pain on exertion**: aortic stenosis, HOCM, arrhythmia, pulmonary hypertension.
- **Fever + murmur + embolic signs**: endocarditis.
- **Hypertension in a child**: coarctation, renal disease (AGN), phaeochromocytoma/neuroblastoma.

Thinking like a clinician

- **Is the child sick or well?** A thriving, pink, playful child with a soft systolic murmur that changes with posture is very likely innocent. A sweaty, thin, breathless infant has heart disease until proven otherwise, even if the murmur is soft (a **large VSD** may be quiet because pressures equalise).
- **Is there cyanosis?** If SpO_2 is low and does not rise with oxygen, think cyanotic CHD (TOF, TGA, tricuspid atresia, TAPVD, truncus). If it rises well, think lung disease.
- **Is there heart failure?** Left-to-right shunts (VSD, AVSD, PDA) cause heart failure at **4–8 weeks**, as pulmonary vascular resistance falls. Duct-dependent left heart lesions (critical coarctation, critical AS, hypoplastic left heart) cause **shock in the first 1–2 weeks**, when the duct closes.
- **Where is the murmur and what does it sound like?** Use the classic-lesions table (section 6).

- **Is something else driving it? Fever, anaemia, sepsis, thyrotoxicosis** cause flow murmurs. Re-examine when the child is well and the Hb is corrected.
- **No murmur does not mean no heart disease**: large VSDs, TGA and coarctation can have little or no murmur; check femorals, SpO_2 and signs of failure.

Common student mistakes

Not feeling femorals; calling infant crackles "heart failure" when it is bronchiolitis; missing cyanosis in an anaemic child; listening only at the front; auscultating a crying baby and calling it "no murmur heard"; looking for ankle oedema and a JVP in an infant; using a cuff that is too small.

8. Complications and Troubleshooting

Problems during the examination

Problem	What to do
Baby cries during auscultation	Settle, feed or let the baby suck a finger; listen again; never record "no murmur" from a crying baby
Breathless or cyanosed infant becomes distressed when laid flat	Keep upright on the parent's lap; distress can trigger a hypercyanotic spell
BP reading high	Repeat with a cuff of the correct size (bladder width ~40% of arm circumference)
Blue fingers in a cold neonate	Peripheral cyanosis; look at the tongue and measure SpO_2
Cyanosis hard to see (anaemia, dark skin)	Pulse oximeter in the right hand and a foot
Soft murmur in a febrile or anaemic child	Treat fever/anaemia and re-examine when well
Liver edge palpable in a wheezy infant	Percuss the upper border and measure the span: hyperinflation pushes a normal liver down

Acting on what you find (management)

A. A murmur in a well child

1. Use the **7 S's** and look for red flags.
2. If **clearly innocent** (well, thriving, normal pulses, S2 and SpO_2, typical innocent murmur): **explain** to the parents that it is a normal sound of blood flow in a healthy heart, that it needs **no treatment, no activity restriction and no antibiotics**, and that it may come and go (louder with fever). Re-examine when well if first heard during fever.
3. If **any doubt**: **ECG, CXR and echo** (refer to a paediatric cardiology clinic or the Uganda Heart Institute).

B. Heart failure in an infant or child

Step	Action	Dose
1	Sit up (infant semi-upright), oxygen if RR ≥70, distress, cyanosis or SpO_2 <90% (WHO)	Aim SpO_2 >90% (careful in duct-dependent lesions)
2	**Furosemide**	**1 mg/kg IV** (or oral); if no diuresis in 2 h, **2 mg/kg**, then **1–2 mg/kg once or twice daily** (WHO Pocket Book)
3	**Spironolactone** (with longer-term diuretics)	**1–3 mg/kg/day** in 1–2 doses (UCG 1.5–3 mg/kg/day)

Step	Action	Dose
4	**Potassium** if furosemide alone for >5 days or with digoxin	**3–5 mmol/kg/day** oral (WHO)
5	**ACE inhibitor** (large left-to-right shunts, MR/AR, cardiomyopathy; specialist)	**Captopril 0.1–0.3 mg/kg/dose 8-hourly** (start low, titrate) or **enalapril 0.05–0.1 mg/kg/day** titrated to ~0.5 mg/kg/day (max 20 mg)
6	**Digoxin** (poor ventricular function, fast AF; specialist)	Load **15 µg/kg** oral, then **5 µg/kg 12-hourly** from 6 h later (max 250 µg/dose) (WHO Pocket Book)
7	**Nutrition**: high-energy feeds (fortified breast milk or formula, e.g. 100–150 kcal/kg/day), small frequent feeds, **NG feeding** if tiring	Do not over-restrict fluids in infants; energy matters for growth
8	Treat **anaemia** (packed cells slowly + furosemide if Hb very low), **infection**, **fever**	—
9	Avoid **IV fluids** where possible; if needed, give carefully calculated volumes	—
10	**Refer** for echo and surgical assessment (UHI)	—

UGANDA CONTEXT

Spironolactone: UCG 2023 gives **1.5–3 mg/kg/day**; other paediatric references give **1–3 mg/kg/day**. Use the UCG range on Ugandan wards. Furosemide and digoxin doses above follow the WHO Pocket Book.

C. Hypercyanotic ("tet") spell

1. **Calm** the child; hold in the **knee-chest position** (older children squat) to raise systemic vascular resistance and push blood through the lungs.
2. **Oxygen**.
3. **Morphine 0.1–0.2 mg/kg** IV, IM or SC (reduces distress and hyperpnoea).
4. **IV fluid bolus 10–20 mL/kg** normal saline (improves RV filling).
5. If persisting: **propranolol 0.015–0.02 mg/kg IV slowly** (some protocols up to 0.1 mg/kg, specialist), **sodium bicarbonate 1–2 mmol/kg** for acidosis, **phenylephrine** (specialist), and anaesthesia/ventilation in refractory spells.
6. **Prevention**: treat **iron deficiency/anaemia**, avoid dehydration, **oral propranolol 0.5–1 mg/kg 6–8-hourly** while awaiting surgery (specialist), and **urgent referral** for surgical repair or a shunt.

D. Suspected duct-dependent lesion in a neonate

- Signs: **cyanosis not improving with oxygen** (duct-dependent pulmonary circulation, e.g. pulmonary atresia; or TGA), or **shock with weak femorals in the first 2 weeks** (duct-dependent systemic circulation: critical coarctation, critical AS, hypoplastic left heart).
- **Prostaglandin** keeps the duct open: **alprostadil (PGE1) IV infusion 5–10 ng/kg/min** (up to 50–100 ng/kg/min in collapse; watch for **apnoea**, fever, hypotension), or **oral dinoprostone (PGE2)** where IV PGE1 is unavailable. Availability in Uganda is limited to referral centres Verify *current availability of alprostadil/dinoprostone in Ugandan referral hospitals*.
- Avoid excessive oxygen in duct-dependent systemic lesions (oxygen promotes duct closure and lowers pulmonary resistance, stealing systemic flow); aim SpO_2 about 75–85% once on prostaglandin (specialist).
- **Urgent referral** to the **Uganda Heart Institute** after discussion.

E. Other common actions

- **Rheumatic fever/RHD** found on examination: start **benzathine penicillin** prophylaxis and refer for echo (see the rheumatic fever topic).
- **Endocarditis prevention**: **good dental hygiene for all children with heart disease**; antibiotic prophylaxis before dental procedures only for highest-risk lesions (prosthetic valves, previous endocarditis, unrepaired cyanotic CHD, first 6 months after repair with prosthetic material): **amoxicillin 50 mg/kg (max 2 g)** orally 30–60 min before.
- **Hypertension**: repeat with the correct cuff on 3 occasions; look for **coarctation** (femorals, 4-limb BP) and **renal disease** (urine dipstick, creatinine, ultrasound).
- **Immunisation**: complete UNEPI vaccines including PCV; influenza and RSV prevention where available for haemodynamically significant CHD.
- **Counselling**: explain the lesion with a simple drawing, the plan (medicines, feeding, follow-up), danger signs (breathing, colour, feeding, sweating), and the referral pathway.

9. Documentation and Presenting Findings

Present in this order: **general → hands → pulses → face/neck → precordium → auscultation → heart failure signs → summary and differential**.

Model documentation (normal heart with an innocent murmur): "Comfortable, pink, HR 112 regular, normal volume, femorals present and equal to brachials, RR 28, SpO_2 98% right hand and 98% foot, BP 92/56 right arm, apex 4th ICS just lateral to MCL, normal character, no heaves or thrills, HS I + II with normal physiological splitting, grade 2/6 vibratory systolic murmur at the LLSE, louder supine, no radiation, liver 1 cm, no oedema, weight-for-age Z 0.2."

Model one-liner: "A 3-month-old girl, 4.1 kg (WAZ −2.6, falling from 0 at birth), with sweating and breathlessness during feeds, RR 64, HR 168, SpO_2 96%, a harsh grade 3/6 pansystolic murmur at the lower left sternal edge with a thrill, a soft apical mid-diastolic flow murmur, a loud P2 and liver 4 cm below the costal margin: **large VSD with heart failure and failure to thrive, possible early pulmonary hypertension**."

Model case presentation: "**Baby Nankya is a 10-week-old girl weighing 3.9 kg** (birth weight 3.2 kg; weight-for-age Z −2.4, length-for-age Z −0.8, weight-for-length Z −2.1), **HIV-exposed** (DNA PCR negative at 6 weeks; on cotrimoxazole prophylaxis, mother virally suppressed on TLD), exclusively breastfed, and has received **BCG, OPV0, and the 6-week UNEPI vaccines** (OPV1, pentavalent 1, PCV1, rotavirus 1). She presents with **3 weeks of fast breathing, sweating with feeds and feeds lasting 40 minutes**. On examination she is thin and pink, **RR 66 with subcostal recession**, **HR 172**, **SpO_2 97% in the right hand and 97% in the foot**, femorals present and equal to brachials, CRT 2 s. The **apex is in the 4th space in the anterior axillary line, thrusting**. There is a **grade 3/6 harsh pansystolic murmur at the lower left sternal edge with a thrill**, an apical mid-diastolic murmur and a **loud P2**. The **liver is 4 cm** below the costal margin, chest clear, no oedema. My impression is a **large ventricular septal defect with heart failure and failure to thrive**, possibly with early pulmonary hypertension. Plan: **furosemide 1 mg/kg twice daily**, spironolactone, high-energy feeds (consider NG), Hb check, **echo** and referral to the Uganda Heart Institute for surgical assessment."

10. OSCE Checklist

Step	Done well when	Critical?
Introduces self, confirms identity, explains, gains consent (and child's assent), offers chaperone	Parent's relationship confirmed	No
Washes hands, warms hands and stethoscope	Before touching the child	No

Step	Done well when	Critical?
Positions appropriately for age, exposes chest respectfully	Lap or 45°; vest kept on older girls until auscultation	No
General inspection	Well/unwell, work of breathing, colour (cyanosis, pallor), dysmorphism, scars, nutrition stated	Yes
Comments on / plots **growth**	Weight, length, head circumference with Z-scores	Yes
Hands	CRT, temperature, clubbing, splinters, peripheral cyanosis, pallor	No
Pulse	Rate (full minute), rhythm, volume, character; brachial in infants	Yes
Femoral pulses	Felt both sides and compared with brachial/radial (radio-femoral delay)	Yes
States intention to measure **BP with the correct cuff**	Right arm ± leg; cuff size explained	Yes
Face/mouth	Central cyanosis (tongue), conjunctival pallor, dentition	No
Neck	JVP (older child), carotid/suprasternal thrill	No
Precordium inspection	Bulge, visible pulsation, scars	No
Apex beat	Position (space, relation to MCL) and character	Yes
Heaves and thrills	Parasternal, all areas, suprasternal	No
Auscultation in all 4 areas	Diaphragm and bell, timed with a pulse	Yes
Listens beyond the 4 areas	**Axilla, below the left clavicle, neck and back**	Yes
Comments on heart sounds	**S1, S2 (splitting)**, added sounds	No
Describes any murmur	Timing, site, grade, radiation, character, posture change	Yes
Uses **manoeuvres** where relevant	Left lateral, sitting forward	No
Lung bases auscultated		No
Liver and **spleen**	Edge and span	Yes
Oedema	Sacral, periorbital, legs	No
States completion steps	**SpO_2 right hand + foot**, urine dipstick, fundoscopy, temperature	Yes
Thanks parent/child, covers, washes hands		No
Summarises	Findings, differential and next investigations (ECG, CXR, echo)	Yes

11. Key Points

High-yield points

- Listen while the baby is quiet; do BP, femorals and liver last if they upset the child.
- Always feel the femorals and measure SpO_2 in the right hand and a foot.
- Infant heart failure = fast breathing, sweating with feeds, poor weight gain, tachycardia, gallop and a big liver.
- Central cyanosis is seen on the tongue; use an oximeter, especially with anaemia or dark skin.
- Innocent murmurs follow the 7 S's in a well, thriving child with normal pulses, S2 and SpO_2.

- Any diastolic murmur, a thrill, an abnormal S2, symptoms or a syndrome makes a murmur pathological.
- A 4/6 murmur has a thrill.
- Apex: 4th space under 5 years, 5th space after.
- Left-to-right shunts present at 4–8 weeks; duct-dependent lesions in the first 1–2 weeks.
- No murmur does not exclude heart disease.
- Any murmur not clearly innocent needs ECG, CXR and echo.

Numbers & doses

Item	Value
HR newborn / infant / 2–5 y / 5–12 y / >12 y	110–160 / 100–160 / 80–140 / 70–120 / 60–100
WHO heart-failure tachycardia	>160 (<12 mo); >120 (1–5 y)
WHO fast breathing	≥60 (<2 mo); ≥50 (2–12 mo); ≥40 (1–5 y)
Lower limit SBP (1–10 y)	70 + 2 × age (years)
Normal CRT	≤2 s (abnormal >3 s in ETAT)
BP cuff	Bladder width ~40% of arm circumference; length 80–100%
Arm–leg BP gradient abnormal	>10–20 mmHg
Hypertension	≥95th centile on 3 occasions (≥130/80 from 13 y)
Newborn SpO_2 screen	Pass ≥95% and difference ≤3%; fail <90%
Murmur with a thrill	≥4/6
Liver edge normal in infants	Up to 2 cm below costal margin
CHD birth prevalence	~8–9 per 1,000 live births
Sibling recurrence of CHD	2–5%
Furosemide in heart failure	1 mg/kg, then 2 mg/kg if no response (WHO)
Spironolactone	1–3 mg/kg/day (UCG 1.5–3)
Potassium supplement	3–5 mmol/kg/day
Captopril	0.1–0.3 mg/kg/dose 8-hourly
Enalapril	0.05–0.1 mg/kg/day, titrated to ~0.5 mg/kg/day (max 20 mg)
Digoxin (WHO)	Load 15 µg/kg; 5 µg/kg 12-hourly (max 250 µg/dose)
Morphine for tet spell	0.1–0.2 mg/kg IV/IM/SC
Fluid bolus for tet spell	10–20 mL/kg NS
Propranolol (tet spell)	0.015–0.02 mg/kg IV slowly; oral 0.5–1 mg/kg 6–8-hourly for prevention
Sodium bicarbonate (tet spell acidosis)	1–2 mmol/kg
Alprostadil (PGE1)	5–10 ng/kg/min (watch for apnoea)
Amoxicillin IE prophylaxis	50 mg/kg (max 2 g) 30–60 min before

Exam traps

- **Trap**: "No murmur, so no heart disease." **Right**: large VSDs, TGA and coarctation can have little or no murmur; check femorals, SpO_2 and signs of failure.
- **Trap**: Looking for ankle oedema and a JVP in an infant. **Right**: look for **hepatomegaly**, tachypnoea, sweating with feeds and poor weight gain.

- **Trap**: Calling central cyanosis by looking at the fingers. **Right**: look at the **tongue** and use a **pulse oximeter**; peripheral cyanosis is common in cold neonates.
- **Trap**: Labelling a soft systolic murmur in a febrile, anaemic child as heart disease. **Right**: correct fever/anaemia and re-examine.
- **Trap**: Measuring BP with a cuff that is too small. **Right**: small cuffs over-read; choose by arm circumference.
- **Trap**: Giving a large fluid bolus to a child in heart failure. **Right**: diuretics, oxygen, sit up; fluids only if clearly hypovolaemic.
- **Trap**: Giving lots of oxygen to a neonate with suspected duct-dependent systemic circulation. **Right**: discuss with a senior; oxygen can close the duct.

Questions seniors ask

Viva questions

1. **Why do left-to-right shunts present with heart failure at 4–8 weeks, not at birth?** Pulmonary vascular resistance is high at birth and falls over the first weeks, so shunt flow (and pulmonary overcirculation) increases.
2. **Why feel the femoral pulses?** To detect coarctation of the aorta (weak or delayed femorals, arm hypertension), which may have no murmur.
3. **What makes a murmur innocent?** The 7 S's in an asymptomatic, thriving child with normal pulses, S2 and SpO_2.
4. **How do you tell cyanotic heart disease from lung disease in a neonate?** Response to oxygen (hyperoxia test), pre/post-ductal saturations, CXR (lung fields), and echo.
5. **What does a fixed split S2 mean?** An atrial septal defect: equalised respiratory variation of right ventricular volume.
6. **Why does the murmur get softer during a tet spell?** Less blood crosses the obstructed RV outflow tract; more shunts right-to-left across the VSD.

On the ward round

1. **Describe the murmur.** Timing, site, grade, radiation, character, change with posture, plus S2 and pulses.
2. **Is this murmur innocent?** Apply the 7 S's and the red flags; if any doubt, echo.
3. **What are the signs of heart failure in this infant?** Tachypnoea, tachycardia, sweating with feeds, poor weight gain, gallop, hepatomegaly.
4. **Why are the femorals important?** Coarctation; duct-dependent systemic circulation in neonates.
5. **What is the significance of a loud P2?** Pulmonary hypertension (e.g. large VSD, AVSD, PDA, MS); urgent referral before irreversible pulmonary vascular disease (Eisenmenger).
6. **How do you manage a tet spell?** Knee-chest, oxygen, morphine 0.1–0.2 mg/kg, fluid bolus 10–20 mL/kg, propranolol, and urgent referral.
7. **Where would you refer a child with suspected structural heart disease in Uganda?** For echo at an RRH cardiac clinic or the **Uganda Heart Institute (Mulago)**, which provides catheter interventions and paediatric cardiac surgery.

Mnemonics

- **Innocent murmur "7 S's"**: Sensitive, Short, Single, Small, Soft, Sweet, Systolic.
- **Describe a murmur "SCRIPT"**: **S**ite, **C**haracter (quality/pitch), **R**adiation, **I**ntensity (grade), **P**itch, **T**iming (and manoeuvres).
- **Heart failure in infants "SWEAT, FAST, BIG LIVER"**: sweating with feeds, fast breathing/fast heart, poor gain, big liver.

- **Cyanotic CHD "5 Ts"**: **T**etralogy of Fallot, **T**ransposition of the great arteries, **T**ricuspid atresia, **T**otal anomalous pulmonary venous drainage, **T**runcus arteriosus.
- **Tet spell treatment "knees to chest, oxygen, morphine, fluids, propranolol"**.
- **Apex "4 under 5"**: 4th space under age 5, 5th space after.

OSCE tips

- **Narrate** as you go ("I am feeling both femoral pulses and comparing them with the right brachial").
- If the baby is quiet at the start, **auscultate first** and say why.
- Always end by stating **completion steps**: SpO_2 right hand and foot, BP, growth chart, urine dipstick, and ECG/CXR/echo if indicated.
- Present findings in order: **general → hands → pulses → face/neck → precordium → auscultation → heart failure signs → summary and differential**.

12. References

1. World Health Organization. Pocket Book of Hospital Care for Children, 2nd edition (section 4.8 Heart failure; Annex 2 drug doses). Geneva: WHO; 2013. https://www.who.int/publications/i/item/978-92-4-154837-3
2. Ministry of Health Uganda. Uganda Clinical Guidelines 2023: section 4.1.3 Heart Failure. Kampala: MoH; 2023. https://library.health.go.ug/uganda-clinical-guidelines-2023
3. Fleming S, Thompson M, Stevens R, et al. Normal ranges of heart rate and respiratory rate in children from birth to 18 years of age: a systematic review of observational studies. Lancet 2011;377:1011–8. https://pubmed.ncbi.nlm.nih.gov/21411136/
4. Heart murmurs in children: evaluation and management. American Family Physician 2022. https://pubmed.ncbi.nlm.nih.gov/35289571/
5. Gewitz MH, et al. Revision of the Jones Criteria for the diagnosis of acute rheumatic fever in the era of Doppler echocardiography: a scientific statement from the American Heart Association. Circulation 2015. https://pubmed.ncbi.nlm.nih.gov/25908771/
6. World Health Organization. WHO guideline on the prevention and diagnosis of rheumatic fever and rheumatic heart disease. Geneva: WHO; 2024. https://www.who.int/publications/i/item/9789240100077
7. Glynn M, Drake WM, editors. Hutchison's Clinical Methods: An Integrated Approach to Clinical Practice. 24th edition. Edinburgh: Elsevier; 2018.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Normal heart rate ranges by age?	Newborn **110–160**; infant **100–160**; 2–5 y **80–140**; 5–12 y **70–120**; >12 y **60–100** /min (approximate).
2	WHO thresholds for tachycardia in heart failure?	**>160/min under 12 months**; **>120/min at 1–5 years** (WHO Pocket Book).
3	WHO fast-breathing thresholds?	**≥60/min under 2 months**; **≥50 at 2–12 months**; **≥40 at 1–5 years**.
4	Lower limit of normal systolic BP (PALS)?	Term neonate **<60**; infant **<70**; 1–10 years **<70 + 2 × age**; ≥10 years **<90** mmHg.
5	How do you choose a BP cuff for a child?	Bladder **width ~40% of mid-upper-arm circumference** and **length 80–100%** of the circumference. Too small a cuff **over-reads**.
6	Why must you always feel the femoral pulses?	Weak, delayed or absent femorals = **coarctation of the aorta** (or interrupted arch, or duct-dependent left heart in a shocked neonate). Check arm vs leg BP (gradient >10–20 mmHg abnormal).
7	Where is the apex beat in children?	**4th intercostal space just lateral to the midclavicular line** under about 5–7 years; **5th space at the midclavicular line** in older children.
8	Heaving vs thrusting apex?	**Heaving** (sustained, not displaced) = **pressure load** (AS, coarctation, hypertension). **Thrusting/displaced** = **volume load** (VSD, PDA, MR, AR, cardiomyopathy).
9	What does a left parasternal heave indicate?	**Right ventricular hypertrophy**: pulmonary hypertension, pulmonary stenosis, TOF, large left-to-right shunts.
10	Levine grading of systolic murmurs?	1 very soft; 2 soft, easily heard; 3 loud, **no thrill**; **4 with thrill**; 5 heard with the stethoscope **edge**; 6 heard with the stethoscope **off the chest**.
11	The 7 S's of an innocent murmur?	**S**ensitive (posture/breathing), **S**hort, **S**ingle (no click), **S**mall (localised), **S**oft (≤2/6), **S**weet (not harsh), **S**ystolic. Plus: asymptomatic, normal pulses, S2, SpO_2 and growth.
12	Still's murmur: features?	Age **2–7 y**; short **vibratory/musical** systolic murmur at the **lower left sternal edge/apex**; louder **lying down** and with fever; softer standing. Innocent.
13	Venous hum: features?	**Continuous** humming murmur **above or below the clavicle** (usually right); **disappears lying flat, turning the head or compressing the jugular vein**. Innocent.
14	Peripheral pulmonary branch stenosis murmur?	Newborn (often preterm); soft ejection systolic at the upper left sternal edge **radiating to both axillae and the back**; resolves by about **6 months**. Innocent.
15	Red flags that a murmur is pathological?	**Diastolic** or **pansystolic**; **grade ≥3 or thrill**; harsh; **click**; **abnormal S2** (fixed split, single, loud P2); symptoms (FTT, sweating, breathlessness, cyanosis, syncope); abnormal pulses/SpO_2; hepatomegaly; syndrome; neonate.
16	VSD murmur?	**Harsh pansystolic** at the **lower left sternal edge**, often with a thrill. Small VSD = loud; large VSD may be softer with an apical mid-diastolic flow murmur and loud P2.

#	Question	Answer
17	ASD signs?	Soft **ejection systolic** murmur at the **upper left sternal edge** (pulmonary flow) and a **fixed, widely split S2**.
18	PDA signs?	**Continuous machinery murmur** in the **left infraclavicular** area, **bounding pulses**, wide pulse pressure. Preterm babies, congenital rubella.
19	Aortic vs pulmonary stenosis murmur?	**AS**: ESM **upper right sternal edge → carotids**, suprasternal thrill, ejection click. **PS**: ESM **upper left sternal edge → back**, ejection click, RV heave.
20	Signs of heart failure in an infant?	**Tachypnoea**, **tachycardia**, **sweating and tiring with feeds**, **poor weight gain**, **gallop**, **hepatomegaly** (>2 cm, tender); oedema (periorbital/sacral) is late; JVP hard to assess.
21	Newborn pulse-oximetry screening thresholds (AAP 2020)?	**Pass**: ≥95% in right hand or foot and difference ≤3%. **Fail**: <90% in either. **Repeat once**: 90–94% or difference >3%.
22	Syndromes and their typical heart lesions?	**Down**: AVSD; **Turner**: coarctation, bicuspid AV; **Noonan**: PS, HCM; **Williams**: supravalvular AS; **DiGeorge**: TOF/truncus/interrupted arch; **Marfan**: aortic root; **rubella**: PDA, peripheral PS.
23	Why do large left-to-right shunts cause heart failure at 4–8 weeks?	**Pulmonary vascular resistance falls** over the first weeks after birth, increasing shunt flow and pulmonary overcirculation.
24	Management of a hypercyanotic (tet) spell?	Calm, **knee-chest position**, **oxygen**, **morphine 0.1–0.2 mg/kg** IV/IM/SC, **NS 10–20 mL/kg**, then propranolol IV (specialist), bicarbonate for acidosis, phenylephrine; urgent referral.
25	The 5 T's of cyanotic congenital heart disease?	**Tetralogy of Fallot, Transposition of the great arteries, Tricuspid atresia, Total anomalous pulmonary venous drainage, Truncus arteriosus.**
26	Where must you listen besides the four classic areas?	**Left axilla** (MR), **left infraclavicular** (PDA), **neck/carotids** (AS), **back between the scapulae** (coarctation, PS, branch PS, PDA).
27	Completion steps after a paediatric CVS exam?	**SpO_2 right hand + foot**, **BP** (± leg/4-limb), **plot growth**, liver span, lung bases, oedema, urine dipstick, temperature, fundoscopy; then **ECG, CXR, echo** if murmur not clearly innocent.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 4-year-old boy (16 kg) is seen with malaria. Temperature 39.0 °C, HR 140. You hear a grade 2/6 short, vibratory systolic murmur at the lower left sternal edge that is louder lying down and softer sitting up. Pulses, S2 and SpO_2 (98%) are normal and he is growing along the 50th centile. What is the MOST appropriate plan?

- A Urgent echo and start furosemide
- B Start benzathine penicillin prophylaxis for rheumatic heart disease
- C Refer to the Uganda Heart Institute for surgery
- D Restrict physical activity until an echo is done
- E Treat the malaria and fever, explain that this is probably an innocent (Still's) murmur, and re-examine when he is well

2. A 2-week-old boy (3.4 kg) is brought grey, breathing fast and feeding poorly. HR 190, CRT 5 s, SpO_2 95% right hand and 82% foot. Brachial pulses are palpable but femoral pulses cannot be felt. Glucose 3.8 mmol/L. What is the MOST likely diagnosis? Advanced

- A Neonatal sepsis only
- B Critical coarctation of the aorta (duct-dependent systemic circulation) presenting as the duct closes
- C Tetralogy of Fallot
- D Innocent peripheral pulmonary stenosis
- E Large atrial septal defect

3. A 9-week-old girl (3.8 kg; birth weight 3.3 kg) takes 45 minutes to breastfeed, sweats on her forehead during feeds and breathes fast. RR 68, HR 170, SpO_2 97%, harsh grade 3/6 pansystolic murmur at the lower left sternal edge with a thrill, liver 4 cm. Chest is clear. What is the MOST likely diagnosis?

- A Bronchiolitis
- B Innocent murmur with poor breastfeeding technique
- C Atrial septal defect
- D Large ventricular septal defect with heart failure
- E Tetralogy of Fallot

4. You are examining an 8-month-old with suspected heart disease. Which sign is the MOST reliable indicator of right-sided heart failure at this age?

- A An enlarged, tender liver that is increasing in size
- B Raised jugular venous pressure
- C Ankle oedema
- D Basal crackles
- E A loud first heart sound

5. A 12-year-old girl (35 kg) has BP 148/92 in the right arm with an appropriately sized cuff on three occasions. Femoral pulses are weak and delayed compared with the radial. There is an ejection systolic murmur heard best between the scapulae. What is the MOST appropriate next step?

- A Start amlodipine and review in 3 months
- B Reassure: BP is normal for age
- C Measure leg BP (4-limb BP), arrange echocardiography and refer for suspected coarctation of the aorta
- D Repeat BP with a smaller cuff
- E Start furosemide for fluid overload

6. A 3-year-old boy with known tetralogy of Fallot awaiting surgery becomes irritable, breathes deeply and quickly, and turns deeply blue after crying. His usual murmur is now very soft. SpO_2 58%. What is the FIRST action?

A Lie him flat and give IV furosemide
B Calm him and hold him in the knee-chest position with oxygen, then give morphine 0.1–0.2 mg/kg
C Give a digoxin loading dose
D Start an IV salbutamol infusion
E Give IV adrenaline 0.01 mg/kg

7. A 6-year-old is seen in clinic with a continuous humming murmur below the right clavicle while sitting. It disappears when she lies flat and when you press gently on the right side of her neck. She is well and growing normally. What is this?

A Patent ductus arteriosus
B Aortic regurgitation
C Arteriovenous fistula
D Pericardial friction rub
E Venous hum (innocent)

8. A 2-year-old (11 kg) is pale, with Hb 4.8 g/dL from severe malaria. HR 168, RR 52 with deep breathing, a soft grade 2/6 ejection systolic murmur at the left sternal edge, a gallop and liver 4 cm. What is the BEST interpretation of the cardiovascular findings?

A Congenital VSD needing urgent echo before any treatment
B Rheumatic carditis needing aspirin
C High-output heart failure from severe anaemia with a flow murmur; treat with transfusion of packed cells (10 mL/kg over 3–4 h) with furosemide 1 mg/kg and re-examine the heart after recovery
D Innocent Still's murmur; no action needed
E Septic shock needing a 20 mL/kg saline bolus

9. A 2-year-old with Down syndrome is reviewed in clinic. Which statement is MOST appropriate?

A All children with Down syndrome should have an echocardiogram, because AVSD and other CHD are common and may be clinically silent
B Only children with a murmur need echo
C Heart disease in Down syndrome always presents in the first week
D Down syndrome is most associated with coarctation of the aorta
E A normal ECG excludes significant heart disease

10. A 5-year-old girl has a soft ejection systolic murmur at the upper left sternal edge. S2 is widely split and does not vary with breathing. She is asymptomatic and growing well. What is the MOST likely diagnosis?

A Pulmonary flow murmur (innocent)
B Ventricular septal defect
C Patent ductus arteriosus
D Secundum atrial septal defect
E Mitral regurgitation

11. During newborn screening at 30 hours, a baby has SpO_2 92% in the right hand and 91% in the foot. He is pink, feeding well, with no murmur. What should you do NEXT according to the AAP 2020 algorithm? Advanced

A Pass the screen and discharge

B Repeat the measurement once in about 1 hour; if still 90–94% or the difference >3%, treat as a failed screen and arrange echo/evaluation

C Start prostaglandin immediately

D Give oxygen and discharge home on oxygen

E Ignore it because there is no murmur

12. A 10-year-old boy has a loud apical pansystolic murmur radiating to the axilla, a displaced thrusting apex in the 6th intercostal space in the anterior axillary line, and polyarthralgia 2 months ago. Which is the MOST likely lesion?

A Aortic stenosis

B Ventricular septal defect

C Innocent Still's murmur

D Mitral regurgitation from rheumatic heart disease

E Pulmonary stenosis

13. In an OSCE, you are asked to examine the cardiovascular system of a 7-month-old who is calm and sleeping on his mother's lap. What is the BEST approach?

A Wake him and lie him flat on the couch for a formal sequence

B Start with BP measurement and femoral pulses, then auscultate

C Ask the mother to feed him first, then examine in the standard adult order

D Skip auscultation because he is too young

E Inspect, then auscultate first while he is quiet on the lap, and leave the more upsetting parts (BP, femorals, liver palpation) until last

14. A 3-day-old baby in the Mengo neonatal unit is centrally cyanosed with SpO_2 70% in both the right hand and foot. SpO_2 barely changes after 10 minutes of 100% oxygen. There is no respiratory distress and a soft murmur. What is the MOST likely category of problem and the next step? Advanced

A Cyanotic congenital heart disease (e.g. transposition); senior review, prostaglandin if available, and urgent echo/referral to the Uganda Heart Institute

B Pneumonia; start antibiotics and increase oxygen

C Polycythaemia; perform partial exchange

D Peripheral cyanosis from cold; warm the baby

E Normal transition; recheck tomorrow

15. Which statement about measuring the apex beat and heart sounds in children is CORRECT?

A A third heart sound is always pathological in children

B The apex is in the 5th intercostal space at the midclavicular line in a 2-year-old

C Sinus arrhythmia (rate varying with breathing) is normal in children

D A murmur with a thrill is grade 2/6

E Physiological S2 splitting is best heard at the apex

Answers & explanations

1. E · Treat the malaria and fever, explain that this is probably an innocent (Still's) murmur, and re-examine when he is well

E is correct: a soft, vibratory, posture-dependent systolic murmur in a thriving child with normal pulses, S2 and SpO_2 fits a **Still's murmur**, accentuated by fever; re-examine when afebrile. A is wrong: there are no heart failure signs and no red flags. B is wrong: nothing suggests rheumatic fever or RHD. C is wrong: no structural lesion has been shown. D is wrong: innocent murmurs need no restrictions, and unnecessary restriction harms the child.

2. B · Critical coarctation of the aorta (duct-dependent systemic circulation) presenting as the duct closes

B is correct: **shock in week 2** with **absent femorals** and **lower post-ductal saturation** is classic **critical coarctation** (or another duct-dependent left heart lesion) as the arterial duct closes; it needs prostaglandin, cautious resuscitation and urgent referral (sepsis cover is also given). A is wrong: sepsis can cause shock but not a pulse and saturation difference between arm and leg. C is wrong: TOF causes cyanosis with equal pulses, rarely shock at 2 weeks. D is wrong: branch PS is innocent and the baby would be well. E is wrong: ASD is usually asymptomatic in infancy.

3. D · Large ventricular septal defect with heart failure

D is correct: heart failure at about **2 months** (as pulmonary resistance falls) with a **harsh pansystolic LLSE murmur and thrill**, hepatomegaly and failure to thrive is a **large VSD**. A is wrong: bronchiolitis gives wheeze/crackles and coryza, not a thrill or a pansystolic murmur. B is wrong: a thrill (grade ≥4/6 at times), hepatomegaly and poor growth are red flags. C is wrong: ASD rarely causes heart failure in infancy and has a soft ESM with fixed split S2. E is wrong: TOF causes cyanosis and does not usually cause heart failure.

4. A · An enlarged, tender liver that is increasing in size

A is correct: in infants, **hepatomegaly** (edge >2 cm, tender, increasing, with a measured span) is the most reliable sign of systemic venous congestion. B is wrong: the infant neck is too short to assess JVP reliably. C is wrong: oedema is a late sign in infants and is periorbital/sacral rather than ankle. D is wrong: basal crackles in infants are more often from pneumonia or bronchiolitis. E is wrong: a loud S1 is not a sign of heart failure.

5. C · Measure leg BP (4-limb BP), arrange echocardiography and refer for suspected coarctation of the aorta

C is correct: arm hypertension with **weak, delayed femorals** and a murmur at the back is **coarctation of the aorta**; confirm with an arm–leg BP gradient and echo, then refer for catheter or surgical repair. A is wrong: antihypertensives without a diagnosis miss a correctable cause. B is wrong: 148/92 is hypertensive at 12 years. D is wrong: a smaller cuff falsely raises BP further. E is wrong: there is no evidence of fluid overload.

6. B · Calm him and hold him in the knee-chest position with oxygen, then give morphine 0.1–0.2 mg/kg

B is correct: this is a **hypercyanotic (tet) spell**; the softer murmur shows reduced RV outflow flow. Knee-chest position raises systemic vascular resistance, oxygen helps, and morphine reduces distress and hyperpnoea; then a fluid bolus and propranolol if needed. A is wrong: furosemide reduces preload and worsens the spell. C is wrong: digoxin increases contractility and can worsen infundibular obstruction. D is wrong: beta-agonists worsen the spell (propranolol, a beta-blocker, helps). E is wrong: adrenaline increases contractility and heart rate and worsens outflow obstruction; it is for cardiac arrest, not tet spells.

7. E · Venous hum (innocent)

E is correct: a **continuous murmur below/above the clavicle that disappears lying flat or with jugular compression** is a **venous hum**, an innocent murmur. A is wrong: a PDA murmur is left infraclavicular and does not disappear with posture or neck compression. B is wrong: AR is early diastolic at the left sternal edge. C is wrong: an AV fistula is rare and does not vanish with jugular compression. D is wrong: a rub is scratchy and varies with position, not abolished by neck pressure.

8. C · High-output heart failure from severe anaemia with a flow murmur; treat with transfusion of packed cells (10 mL/kg over 3–4 h) with furosemide 1 mg/kg and re-examine the heart after recovery

C is correct: **severe anaemia** causes **high-output heart failure** (tachycardia, gallop, hepatomegaly, acidotic breathing) and a **flow murmur**; transfuse packed cells slowly with furosemide, treat malaria, and re-examine when Hb is corrected. A is wrong: the soft ESM is explained by anaemia; echo can follow if the murmur persists. B is wrong: nothing suggests rheumatic fever at age 2. D is wrong: the child is in heart failure, so action is urgent. E is wrong: a saline bolus worsens anaemic heart failure.

9. A · All children with Down syndrome should have an echocardiogram, because AVSD and other CHD are common and may be clinically silent

A is correct: about half of children with Down syndrome have CHD (classically **AVSD**), which may have little or no murmur, and pulmonary hypertension develops early, so **every child needs an echo**. B is wrong: large AVSDs can be quiet. C is wrong: many present with heart failure at 4–8 weeks or later, or remain silent until pulmonary hypertension. D is wrong: coarctation is linked to Turner syndrome. E is wrong: a normal ECG does not exclude structural disease (though a superior axis suggests AVSD).

10. D · Secundum atrial septal defect

D is correct: a **fixed, widely split S2** with an upper left sternal edge ESM (increased pulmonary flow) is the hallmark of an **ASD**; arrange echo. A is wrong: an innocent pulmonary flow murmur has **normal physiological splitting**. B is wrong: VSD gives a pansystolic LLSE murmur. C is wrong: PDA gives a continuous infraclavicular murmur. E is wrong: MR is apical pansystolic to the axilla.

11. B · Repeat the measurement once in about 1 hour; if still 90–94% or the difference >3%, treat as a failed screen and arrange echo/evaluation

B is correct: a reading of **90–94%** in both limbs is an **indeterminate** result; the AAP 2020 algorithm repeats it once, and a persistent result is a failed screen needing evaluation (exclude infection and lung causes; echo). A is wrong: a pass needs ≥95%. C is wrong: prostaglandin is for suspected duct-dependent disease in a sick or clearly cyanosed baby, after senior review. D is wrong: home oxygen without a diagnosis is unsafe. E is wrong: many critical CHDs have no murmur.

12. D · Mitral regurgitation from rheumatic heart disease

D is correct: an **apical pansystolic murmur to the axilla** with a **displaced, thrusting (volume-loaded) apex** in a school-age Ugandan child after joint pains is **rheumatic MR**; echo and start BPG. A is wrong: AS is an ESM at the upper right sternal edge to the carotids with a heaving, non-displaced apex. B is wrong: VSD is loudest at the lower left sternal edge, not radiating to the axilla. C is wrong: Still's murmurs are soft and vibratory with a normal apex. E is wrong: PS is an upper left sternal edge ESM radiating to the back.

13. E · Inspect, then auscultate first while he is quiet on the lap, and leave the more upsetting parts (BP, femorals, liver palpation) until last

E is correct: paediatric examination is **opportunistic**: listen while the infant is quiet, and do the parts that cause crying (BP, femorals, liver) last, while still completing everything. A is wrong: waking him makes auscultation impossible. B is wrong: starting with upsetting steps ruins auscultation. C is wrong: feeding can help, but a rigid adult order loses the quiet window. D is wrong: auscultation is essential at every age.

14. A · Cyanotic congenital heart disease (e.g. transposition); senior review, prostaglandin if available, and urgent echo/referral to the Uganda Heart Institute

A is correct: **central cyanosis without respiratory distress that does not improve with 100% oxygen** (failed hyperoxia test) points to **cyanotic CHD** such as TGA; keeping the duct open with prostaglandin and urgent cardiology referral are needed. B is wrong: lung disease usually causes respiratory distress and improves with oxygen. C is wrong: polycythaemia can cause plethora and mild desaturation, not SpO_2 70% unresponsive to oxygen. D is wrong: the tongue/SpO_2 show central cyanosis. E is wrong: SpO_2 70% on day 3 is never normal.

15. C · Sinus arrhythmia (rate varying with breathing) is normal in children

C is correct: **sinus arrhythmia** (faster heart rate on inspiration) is normal in children. A is wrong: an S3 is common and normal in healthy children (abnormal as part of a gallop with tachycardia and heart failure). B is wrong: under about 5 years the apex is in the 4th space just lateral to the midclavicular line. D is wrong: a thrill means grade ≥4/6. E is wrong: S2 splitting is best heard at the upper left sternal edge (pulmonary area).

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The baby who sweats when she feeds

Baby Nabirye, 8 weeks old, is brought to Mengo paediatric outpatients because she 'breathes like she has been running' and sweats when breastfeeding. Her mother says feeds take nearly an hour.

1. What focused history will you take?

> Feeding (duration, stopping to breathe, sweating, vomiting), breathing at rest, **weight gain** (child health card), colour changes, cough/fever (pneumonia), pregnancy (diabetes, rubella, drugs, HIV), birth and neonatal history (SpO_2 screen, murmur), **family history of CHD**, immunisation (UNEPI), maternal **HIV status** and infant prophylaxis.

2. Describe how you will examine her and in what order.

> Wash and warm hands; while she is quiet on her mother's lap: **inspect** (breathing, colour, sweating, dysmorphism, nutrition), **count RR and HR** for 60 s, then **auscultate** (4 areas, axilla, back). Then hands (CRT), **brachial and femoral pulses**, precordium (apex, heaves, thrills), **liver**, lung bases, oedema. Finish with **SpO_2 right hand + foot**, BP and **plotting weight/length**.

3. Findings: weight 3.7 kg (birth 3.2 kg; WAZ −2.5), RR 66 with recession, HR 168, SpO_2 96%/96%, femorals normal, apex 4th ICS anterior axillary line, thrusting, harsh 3/6 pansystolic murmur at LLSE with thrill, apical mid-diastolic murmur, loud P2, liver 4 cm. What is your assessment?

> **Large VSD** (left-to-right shunt) with **heart failure** (tachypnoea, tachycardia, hepatomegaly, sweating with feeds) and **failure to thrive**; the **apical mid-diastolic flow murmur** shows a large shunt and the **loud P2** suggests rising pulmonary pressure. Differentials: AVSD (check for Down features), PDA (continuous murmur), coarctation (femorals normal).

4. What investigations will you request?

> **Echocardiogram** (definitive), **CXR** (cardiomegaly, plethoric lungs, exclude pneumonia), **ECG** (biventricular hypertrophy; superior axis if AVSD), **FBC/HemoCue** (anaemia worsens failure), U&E before diuretics, HIV PCR status confirmation.

5. Write her initial management.

> Semi-upright, oxygen if SpO_2 <90% or distress. **Furosemide 1 mg/kg (≈4 mg) once or twice daily** (oral/IV), **spironolactone 1–2 mg/kg/day**; consider **captopril** (0.1 mg/kg/dose 8-hourly, titrated) after specialist advice. **Feeding**: frequent small feeds, expressed breast milk fortified/high-energy, **NG feeds** if tiring (aim ~150 mL/kg/day and 120–150 kcal/kg/day). Treat anaemia and infection; continue immunisations. **Refer to the Uganda Heart Institute** for surgical closure before irreversible pulmonary hypertension.

6. Explain to the mother in plain words.

> "There is a hole between the two pumping chambers of the heart. Too much blood is going to the lungs, so she breathes fast, sweats and uses lots of energy, which is why she is not gaining weight. The medicines remove extra water, and special feeding helps her grow. She needs a heart scan and a heart surgeon to close the hole, ideally in the next few months. Come back at once if she breathes harder, turns blue, or cannot feed."

Take-away: In infants, heart failure shows as fast breathing, sweaty feeds, poor weight gain and a big liver. A harsh LLSE pansystolic murmur with a loud P2 means a large VSD that needs medical treatment,

nutrition and early surgical referral.

Case 2: The murmur heard during malaria

Ssemwogerere, 5 years, is admitted to the ward with uncomplicated malaria. A colleague hears a murmur and asks you whether he has 'a hole in the heart'.

1. What history will help you decide whether the murmur matters?

> Exercise tolerance compared with peers, breathlessness, sweating, **syncope or chest pain on exertion**, blue spells, growth, previous murmurs, **joint pains or chorea** (rheumatic fever), sore throats, family history of CHD or sudden death.

2. He is active, has never been breathless and is growing on the 50th centile. Temperature 38.8 °C, Hb 10.2. What will you examine for?

> General (well, no syndrome), **pulses including femorals**, BP, apex position/character, heaves, thrills; **heart sounds** (S2 splitting at the upper left sternal edge), clicks; **murmur**: timing, site, grade, radiation, character, **change with posture**; liver, lung bases; SpO_2.

3. Findings: grade 2/6 short vibratory systolic murmur at LLSE to the apex, louder lying down, softer standing; normal S2 splitting; no clicks; femorals normal; SpO_2 99%. What is your conclusion?

> **Innocent (Still's) murmur**, accentuated by fever: meets the **7 S's** with no red flags (asymptomatic, normal growth, pulses, S2 and SpO_2).

4. What would have made you worried instead?

> **Diastolic** or **pansystolic** murmur, **grade ≥3/6 or a thrill**, harsh quality, **ejection click**, **fixed split or single loud S2**, weak femorals, hypertension, low SpO_2, hepatomegaly, symptoms, poor growth, or a syndrome. Any of these → ECG, CXR, echo.

5. How will you explain this to the parents?

> "The noise is the sound of blood flowing through a normal heart. It is louder now because of the fever. It is very common in healthy children, needs no medicine, and he can play and go to school normally. We will listen again when he is well."

6. What is your follow-up plan?

> Treat malaria; **re-examine when afebrile** (e.g. at discharge or review). If the murmur changes or any red flag appears, arrange **echo**. Document clearly to avoid repeated unnecessary referrals.

Take-away: Most murmurs heard in febrile or anaemic children are innocent. Use the 7 S's, check femorals, S2 and SpO_2, and reassure confidently when there are no red flags, re-examining once the child is well.

Case 3: The blue toddler who squats

Mutebi, a 2-year-old boy (10 kg), is brought to the emergency unit after becoming 'very blue and floppy' while crying. His mother says he likes to squat after playing and his lips have been dusky since infancy.

1. What is happening and what are your first actions?

> Likely a **hypercyanotic (tet) spell** in **tetralogy of Fallot**. **Calm** the child (keep with mother), **knee-chest position**, **oxygen**, SpO_2 monitoring, and **morphine 0.1–0.2 mg/kg (1–2 mg) IV/IM/SC**. IV access when calmer, **NS 10–20 mL/kg** bolus, check glucose and Hb. If no improvement: **propranolol IV** (specialist), bicarbonate for acidosis; call a senior.

2. After the spell resolves, what will you look for on examination?

> **Central cyanosis**, **clubbing**, growth, dysmorphism (DiGeorge), pulses (normal), RV heave, **ejection systolic murmur at the upper/mid left sternal edge** (RVOT), **single S2**; no heart failure signs usually; SpO_2 (e.g. 75–85%); Hb/Hct (polycythaemia or relative iron deficiency).

3. What do you expect on ECG and CXR?

> **ECG**: right axis deviation, **RVH**. **CXR**: **boot-shaped heart** (upturned apex from RVH, concave pulmonary bay), **oligaemic lung fields**, right aortic arch in some.

4. Why does squatting help and why did the murmur go quiet during the spell?

> Squatting **raises systemic vascular resistance** (kinks femoral arteries), so less blood shunts right-to-left across the VSD and more goes to the lungs. During a spell, RV outflow obstruction worsens, so **less blood crosses the pulmonary outflow** and the ejection murmur softens.

5. What longer-term plan will you set up?

> **Urgent referral to the Uganda Heart Institute** for echo and surgery (complete repair or a Blalock–Taussig shunt). Meanwhile: **treat iron deficiency** (microcytic Hb), avoid dehydration, **oral propranolol** (specialist), teach parents the **knee-chest manoeuvre**, dental hygiene and **endocarditis prophylaxis** before dental procedures (unrepaired cyanotic CHD), complete immunisations.

6. Present him on the ward round.

> "Mutebi, 2 years, 10 kg (WAZ –1.6), immunised per UNEPI, HIV-negative, with cyanosis since infancy and squatting, admitted after a hypercyanotic spell treated with knee-chest positioning, oxygen and morphine. Now SpO_2 80%, clubbed, RV heave, grade 3/6 ESM at the left sternal edge, single S2. Probable tetralogy of Fallot. Plan: echo, CXR, ECG, FBC/ferritin, propranolol after cardiology advice, and urgent UHI referral for surgery."

Take-away: Cyanosis with squatting and a left sternal edge ESM suggests tetralogy of Fallot. Treat spells with knee-chest, oxygen, morphine and fluids, prevent them by treating anaemia and dehydration, and refer for surgery.

References

1. Pocket book of hospital care for children, 2nd edition (section 4.8 Heart failure; Annex 2 drug doses). *World Health Organization*, 2013. https://www.who.int/publications/i/item/978-92-4-154837-3
2. Uganda Clinical Guidelines 2023: section 4.1.3 Heart Failure. *Ministry of Health Uganda*, 2023. https://library.health.go.ug/uganda-clinical-guidelines-2023
3. Normal ranges of heart rate and respiratory rate in children from birth to 18 years of age: a systematic review of observational studies. *Fleming S, et al. Lancet 2011;377:1011-8*, 2011. https://pubmed.ncbi.nlm.nih.gov/21411136/
4. Heart Murmurs in Children: Evaluation and Management. *American Family Physician 2022*, 2022. https://pubmed.ncbi.nlm.nih.gov/35289571/

5. Revision of the Jones Criteria for the diagnosis of acute rheumatic fever in the era of Doppler echocardiography (AHA scientific statement). *Gewitz MH, et al. Circulation 2015*, 2015. https://pubmed.ncbi.nlm.nih.gov/25908771/
6. WHO guideline on the prevention and diagnosis of rheumatic fever and rheumatic heart disease. *World Health Organization*, 2024. https://www.who.int/publications/i/item/9789240100077

A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.